Prediction and Linear Regression

Big Data y Machine Learning para Economía Aplicada

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Agenda

Prediction in the Population

Before any data: what is the best we could do if we knew the full joint distribution?

- We observe random variables $(Y, X) \sim F$.
- A predictor is a function
$$f : \mathcal{X} \to \mathbb{R}.$$
- The population risk under squared loss is
$$R(f) = \mathbb{E}\left[(Y - f(X))^2\right].$$
- The best predictor solves
$$f^* = \arg\min_f R(f).$$

Optimal Predictor Under Squared Loss

- Start from
$$R(f) = \mathbb{E}\left[(Y - f(X))^2\right].$$
- Writing the expectation as an integral,
$$R(f) = \int_{\mathcal{X}} \int_{\mathcal{Y}} (y - f(x))^2 f_{X,Y}(x,y)\, dy\, dx.$$
- Using
$$f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x),$$
- we get
$$R(f) = \int_{\mathcal{X}} \left[\int_{\mathcal{Y}} (y - f(x))^2 f_{Y|X}(y \mid x)\, dy\right] f_X(x)\, dx.$$

Pointwise Minimization

- For each fixed x, minimize
$$\int_{\mathcal{Y}} (y - f(x))^2 f_{Y|X}(y \mid x)\, dy.$$
- The first-order condition is
$$-2 \int_{\mathcal{Y}} (y - f(x)) f_{Y|X}(y \mid x)\, dy = 0.$$
- Therefore,
$$f^*(x) = \int_{\mathcal{Y}} y f_{Y|X}(y \mid x)\, dy = \mathbb{E}[Y \mid X = x].$$
- So,
$$f^*(X) = \mathbb{E}[Y \mid X].$$
- Second-order condition?

From Best Predictor to Linear Regression

- The unrestricted best predictor is

$$f^*(X) = \mathbb{E}[Y \mid X].$$

- But $\mathbb{E}[Y \mid X]$is unknown.

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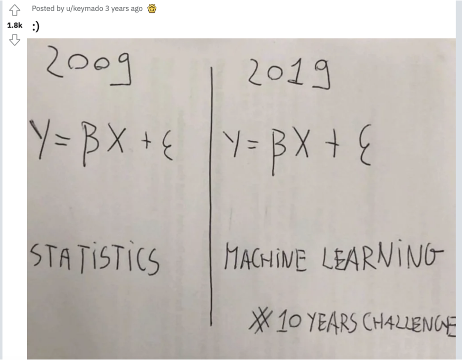

Source: Reddit, u/keymado.

Best Linear Predictor

- Linear regression restricts the predictor to be linear in the parameters:

$$f(X) = X'\beta.$$

- This is still flexible: X may include transformations, interactions, and polynomials.

Best Linear Predictor

- The best linear predictor (β^*) solves

$$\beta^* = \arg\min_{\beta} \mathbb{E}\left[(Y - X'\beta)^2\right].$$

- Thus, linear regression is the best approximation to

$$\mathbb{E}[Y \mid X]$$

within the class of functions that are linear in the parameters.

Population vs Sample

- In the population:
$$\beta^* = \arg\min_{\beta} \mathbb{E}\left[(Y - X'\beta)^2\right].$$
- In the sample:
$$\hat{\beta} = \arg\min_{\beta} \frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i'\beta)^2.$$
- OLS is the sample analogue of the best linear predictor.

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Why Do We Need Statistical Properties?

- In the population, the best linear predictor solves

$$\beta^* = \arg\min_{\beta} \mathbb{E}\left[(Y - X'\beta)^2\right].$$

- In the sample, OLS solves

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i'\beta)^2.$$

- For machine learning, the object of interest is the prediction rule:

$$\hat{f}(x) = x'\hat{\beta}.$$

- The key question is:

How good is $\hat{f}(x)$ as a predictor?

Gauss–Markov Theorem

- Under the classical linear model assumptions,

$$Y = X'\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2,$$

- OLS is BLUE:

$$\textbf{B}\mathit{est} \quad \textbf{L}\mathit{inear} \quad \textbf{U}\mathit{nbiased} \quad \textbf{E}\mathit{stimator}.$$

Predicting the Mean vs Predicting a New Outcome

- There are two different prediction problems.
- **Predicting the conditional mean:**

$$f^*(x) = \mathbb{E}[Y \mid X = x].$$

- OLS estimates it with

$$\hat{f}(x) = x'\hat{\beta}.$$

Predicting the Mean vs Predicting a New Outcome

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- **Predicting the conditional mean:**

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- **Predicting a new realization:**

$$Y_{\text{new}} = f^*(x) + \varepsilon_{\text{new}}.$$

Limitations of Gauss–Markov

- Gauss–Markov is powerful, but limited.
- It requires:
  -
  -
  -
  -

Large Sample Properties

- Under weaker regularity conditions,

$$\hat{\beta} \xrightarrow{p} \beta^*.$$

- That is, OLS is consistent for the population linear projection.
- Also,

$$\sqrt{n}(\hat{\beta} - \beta^*) \xrightarrow{d} N(0, \Gamma).$$

- This asymptotic distribution is the basis for:
  - standard errors,
  - confidence intervals,
  - hypothesis tests.

From Statistical Guarantees to Prediction Quality

- So far: unbiasedness, minimum variance within a class, asymptotic normality.
- All of these are statements about $\hat{\beta}$.
- But our target is the prediction rule $\hat{f}(x) = x'\hat{\beta}$, and what matters for it is expected prediction error.

But Is Unbiasedness What We Really Want?

- Gauss–Markov focuses on unbiasedness and variance.
- But prediction quality depends on expected prediction error:

$$\mathbb{E}\left[(Y - \hat{f}(X))^2\right].$$

Classical inference and the “Big Data Way”

- Inference requires knowing the sampling distribution of $\hat{\beta}$ or $\hat{f}(x)$.
- Classical econometrics derives it analytically:

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Classical inference and the "Big Data Way"

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- Here we are going to take a different route:

Use the data to approximate the sampling distribution.

- This will be useful when analytical formulas are complicated, or when we want uncertainty for more complex prediction methods.

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Numerical Properties

- Numerical properties have nothing to do with how the data was generated
- These properties hold for every data set
- They hold for any n and require no assumption about how u was generated: they are facts of linear algebra, not of probability
- Two matrices, P_X and M_X, generate all of them

Projection

- OLS Residuals:

$$
\begin{aligned}
e &= y - \hat{y} && (1)\\
&= y - X\hat{\beta} && (2)
\end{aligned}
$$

- replacing $\hat{\beta}$

$$
\begin{aligned}
e &= y - X(X'X)^{-1}X'y && (3)\\
&= (I - X(X'X)^{-1}X')y && (4)
\end{aligned}
$$

Frisch-Waugh-Lovell (FWL) Theorem

- Linear Model: $Y = X\beta + u$
- Split it: $Y = X_1\beta_1 + X_2\beta_2 + u$
  - $X = [X_1 \, X_2]$, X is $n \times k$, X_1 $n \times k_1$, X_2 $n \times k_2$, $k = k_1 + k_2$
  - $\beta = [\beta_1 \, \beta_2]$

Theorem

1. The OLS estimates of β_2 from these equations

$$y = X_1\beta_1 + X_2\beta_2 + u \quad (5)$$

$$M_{X_1}y = M_{X_1}X_2\beta_2 + \text{residuals} \quad (6)$$

   are numerically identical

2. the OLS residuals from these regressions are also numerically identical

FWL: Three? Questions, One Tool

FWL splits a regression into pieces and shows what is inside each coefficient. That answers three distinct questions:

1. **Computation.** How do we estimate with very high-dimensional fixed effects without building huge matrices? (Carneiro et al.: 23.4 TB)
2. **Identification.** Which comparisons is the estimator actually making? Where do the weights come from? (Goodman-Bacon 2021)
3. **Diagnostics for Inference and Prediction.** Influential Observations, R^2, LOOCV, etc.

Applications

- Why FWL is useful in the context of big volume of data?
- A computationally inexpensive way of
  - Removing nuisance parameters
    - E.g. the case of multiple fixed effects. The traditional way is either apply the within transformation with respect to the FE with more categories then add one dummy for each category for all the subsequent FE
    - Not feasible in certain instances.

Applications: Fixed Effects

- For example: Carneiro, Guimarães, & Portugal (2012) *AEJ: Macroeconomics*

$$\ln w_{ijft} = x_{it}\beta + \lambda_i + \theta_j + \gamma_f + \delta_t + u_{ijft} \tag{7}$$

  - Data set 31.6 million observations, with 6.4 million individuals (i), 624 thousand firms (f), and 115 thousand occupations (j), 11 years (t).
  - From their paper
    "In our application, we first make use of the Frisch-Waugh-Lovell theorem to remove the influence of the three high- dimensional fixed effects from each individual variable, and, in a second step, implement the final regression using the transformed variables. With a correction to the degrees of freedom, this approach yields the exact least squares solution for the coefficients and standard errors"

Applications: Fixed Effects

American Economic Journal: Economic Policy 2014, 6(3): 63–91
http://dx.doi.org/10.1257/pol.6.3.63

Friends in High Places†

By Lauren Cohen and Christopher J. Malloy*

We demonstrate that personal connections amongst US politicians have a significant impact on Senate voting behavior. Networks based on alumni connections between politicians are consistent predictors of voting behavior. We estimate sharp measures that control for common characteristics of the network, as well as heterogeneous impacts of a common network characteristic across votes. We find that the effect of alumni networks is close to 60 percent as large as the effect of state-level considerations. We show that politicians use school ties as a mechanism to engage in vote trading ("logrolling"), and that alumni networks help facilitate the procurement of discretionary earmarks. (*JEL* D72, D85, Z13)

Source: Cohen & Malloy (2014), AEJ: Economic Policy.

Applications: Fixed Effects

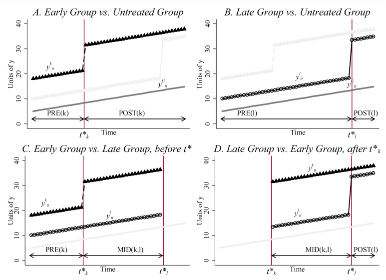
photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

Applications: What is $\hat{\beta}$ made of?

- TWFE

$$y_{it} = \alpha_i + \gamma_t + \beta D_{it} + u_{it} \tag{8}$$

- What happens if adoption is staggered?



Source: Goodman-Bacon (2021), Journal of Econometrics.

Applications: Fixed Effects

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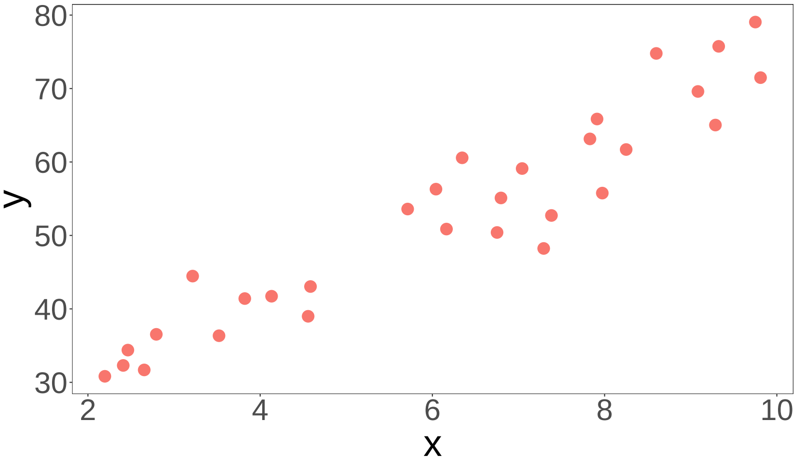
Applications: Influential Observations



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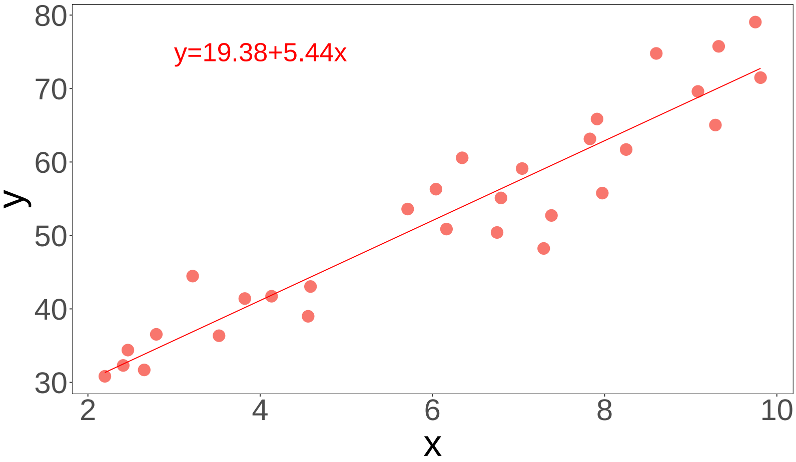

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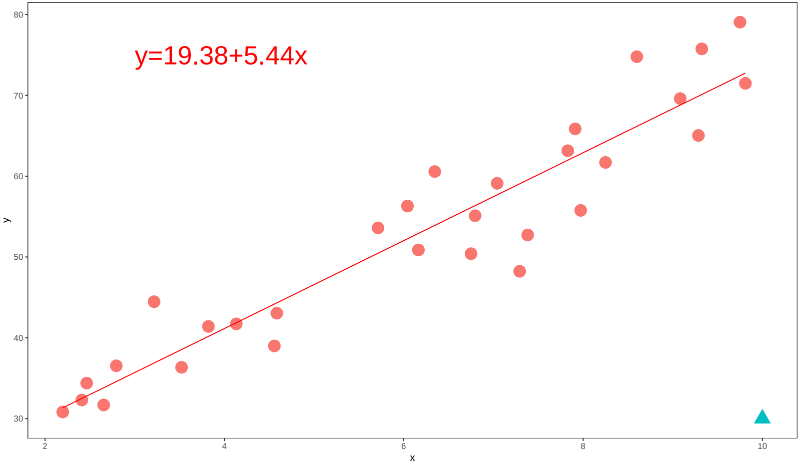
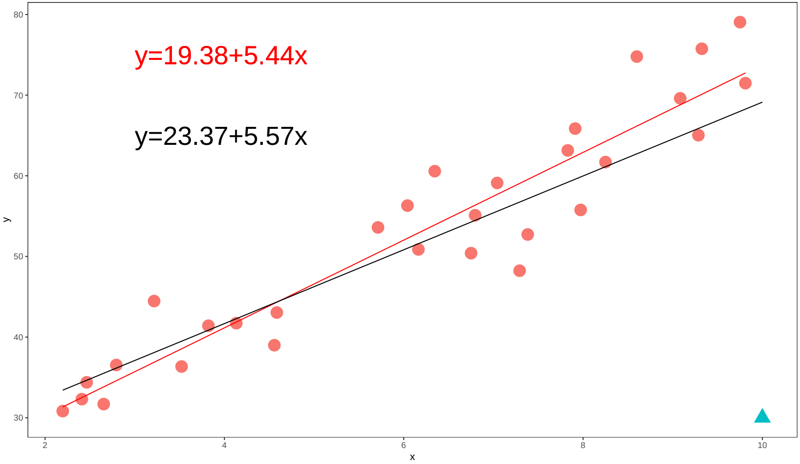


Applications: Influential Observations



Applications: Influential Observations

Consider a dummy variable e_j which is an $n-vector$ with element j equal to 1 and the rest is 0. Include it as a regressor

$$y = X\beta + \alpha e_j + u \tag{9}$$

using FWL we can do

$$M_{e_j} y = M_{e_j} X\beta + r \tag{10}$$

- β and *residuals* from both regressions are identical
- Same estimates as those that would be obtained if we deleted observation j from the sample. We are going to denote this as $\beta^{(j)}$

Note:

- $M_{e_j} = I - e_j(e_j'e_j)^{-1}e_j'$
- $M_{e_j} y = y - e_j(e_j'e_j)^{-1}e_j'y = y - y_j e_j$
- $M_{e_j} X$ is X with the j *row* replaced by zeros

Applications: Influential Observations

$$y = X\beta^{(j)} + \alpha e_j + u \tag{11}$$

$$\tag{12}$$

Let's define a new matrix $Z = [X, e_j]$

$$y = Z\theta + u \tag{13}$$

we can write it as

$$y = P_Z y + M_z y \tag{14}$$

$$y = X\hat{\beta}^{(j)} + \hat{\alpha} e_j + M_Z y \tag{15}$$

Pre-multiplying by P_X and using the fact that $P_X M_Z = 0$

$$P_X y = X\hat{\beta}^{(j)} + \hat{\alpha} P_X e_j \tag{16}$$

$$X\hat{\beta} = X\hat{\beta}^{(j)} + \hat{\alpha} P_X e_j \tag{17}$$

$$X(\hat{\beta} - \beta^{(j)}) = \hat{\alpha} P_X e_j \tag{18}$$

Applications: Influential Observations

How to calculate α? FWL once again

$$M_X y = \hat{\alpha} M_X e_j + res \tag{19}$$

$$\hat{\alpha} = (e_j' M_X e_j)^{-1} e_j' M_X y \tag{20}$$

- $e_j' M_X y$ is the j element of $M_X y$, the vector of residuals from the regression including all observations
- $e_j' M_x e_j$ is just a scalar, the diagonal element of M_X

  Then

$$\hat{\alpha} = \frac{\hat{u}_j}{1 - h_j} \tag{21}$$

where h_j is the j diagonal element of P_X

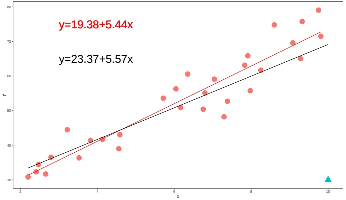
Applications: Influential Observations

Finally we get

$$(\hat{\beta}^{(j)} - \hat{\beta}) = -\frac{1}{1-h_j}(X'X)^{-1}X_j'\hat{u}_j \tag{22}$$

Influence depends on two factors

- $\hat{u}_j$ large residual $\rightarrow$ related to y coordinate
- $\hat{h}_j \rightarrow$ related to x coordinate



HW. case of $y = \alpha + \beta x + u$ (ISLR) App

Recap: One Tool, Three Problems

What did we do today?

- We started from the prediction problem: the best predictor is $\mathbb{E}[Y \mid X]$, and OLS is its best linear approximation.
- Behind $\hat{\beta}$ there are two matrices, P_X and M_X, and one theorem: FWL.
- With that single tool we solved three problems:
  1. **Computation:** very high-dimensional fixed effects without building huge matrices (Carneiro et al.: 23.4 TB $\rightarrow$ feasible).
  2. **Identification:** which comparisons the estimator is actually making, and where the weights come from (Goodman-Bacon).
  3. **Diagnostics:** influential observations

Knowing OLS well pays off, and pays off big: it is not just an estimator, it is a toolbox. The same algebra that computes it tells us what it compares and when to distrust it.